

Module 6: Offer Price Range and Due Diligence

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Offer price range

Assume that the acquirer will buy 100% of the target's equity

Minimum Total Offer Price
acceptable to the **target**

\$100m

Target's
Standalone
Equity value

Merger Gains

Target's
Standalone
Equity Value

Maximum Total Offer Price
acceptable to the **acquirer**

\$50m

100m

\$150m

Acquisition Premium (\$) = The Target's Piece of NPV from the M&A Deal

- Acquisition Premium (\$) = $\frac{\$130m}{\$100m} = \$30m$ Total Offer Price – Target's standalone value of equity

What the target's shareholders
receive from the M&A

What the target's shareholders
give away in the M&A

- Acquisition Premium (\$): the target's piece of NPV (merger gains) from the M&A deal. $\$50m$
- Merger gains (i.e., synergy + other M&A benefits): the total NPV generated the M&A deal.
- Question: what is the acquirer's NPV from the M&A deal?

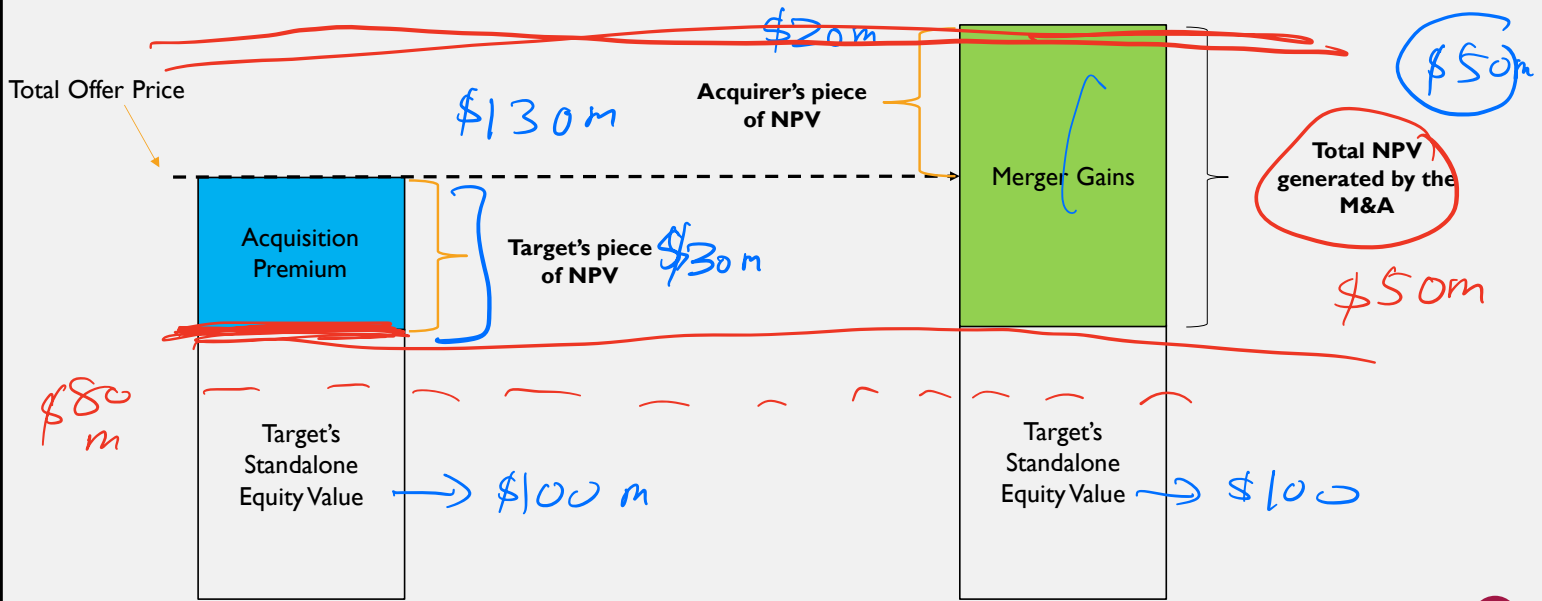
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$\$50m - \$30m$

$= \$20m$

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\$170m Linking M&A concepts with NPVs



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$$7 \approx 45 \text{ PM}$$

$$\text{Offer} = \$80m$$

$$\text{offer} = \$170m$$

$$170m - 100m$$

$$= 70m$$

$$50m - 70m$$

$$\text{Total} \quad \text{St}$$

$$= -20m.$$

A simple example

Assume the target's pre-M&A MV of equity = standalone value of equity

Acquirer: all-equity firm A

- $V_A = \$400$ million $\rightarrow E_A = V_A = 400$
- # shares = 100 million
- Share value = $\$400 \text{ m} / 100 = \4 per share

Target: all-equity firm B

- $V_B = \$200$ million $E_B = 200 \text{ milli..}$
- # shares = 200 million
- Share value = $\$200 / 200 = \1 per share

Combined firm value (with merger gains):

- $V_{AB} = \$700$ million
- Merger Gains = $V_{AB} - (V_A + V_B) = 700 - (400 + 200) = \100 million

DCF

Total Firm Values

DCF

$$V_{AB} - (V_A + V_B)$$

$$V_B^* - V_B$$

What are the merger gains (\$100 million this case)?

- The \$100 million are the total NPV generated by this M&A deal.
- The \$100 million needs to be split between the acquirer and the target.
- Assume that the acquirer intends to acquire the targets' 100% equity with cash.
 - Question #1: What should be the *minimum* offer price per share acceptable to the target? → \$1.
 - Question #2: What should be the *maximum* offer price per share acceptable to the acquirer?

$$E_B = \$200 \text{ million}$$

$$\begin{aligned} & \$200 \text{ million} + \$100 \text{ m} \\ & = \$300 \text{ million} \end{aligned}$$

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$$\frac{300 \text{ million}}{200 \text{ million}} = \$1.5$$

NPV from Acquirer 0

NPV from Tgt

$$\begin{aligned} & \$300 \text{ million} - \$200 \text{ million} \\ & = \$100 \text{ million} \end{aligned}$$

Check Your Understanding

- Assume that the total offer price is \$230 million in cash →
- What is the NPV of this M&A deal from the target's perspective?

$$\begin{array}{ccc}
 230m & - & 200m & = & \$30m \\
 \downarrow & & \downarrow & & \\
 \text{Receive} & & \text{give up} & &
 \end{array}$$

- What is the NPV of this M&A deal from the acquirer's perspective?

$$\$100 - \$30 = \$70m$$

Check Your Understanding

- Assume that the total offer price is \$330 million
- What is the NPV of this M&A deal from the target's perspective?

$$\$330 \text{ million} - \$200 \text{m} = \$130 \text{m}$$

- What is the NPV of this M&A deal from the acquirer's perspective?

$$\$120 \text{m} - \$130 \text{m} = -20 \text{m}$$

The acquirer paid too high offer price, and this deal destroy value for acquirer shareholders!

\$330mln

Minimum Offer Price (\$200 mln)	Maximum Offer Price (\$300mln)
---------------------------------------	--------------------------------------

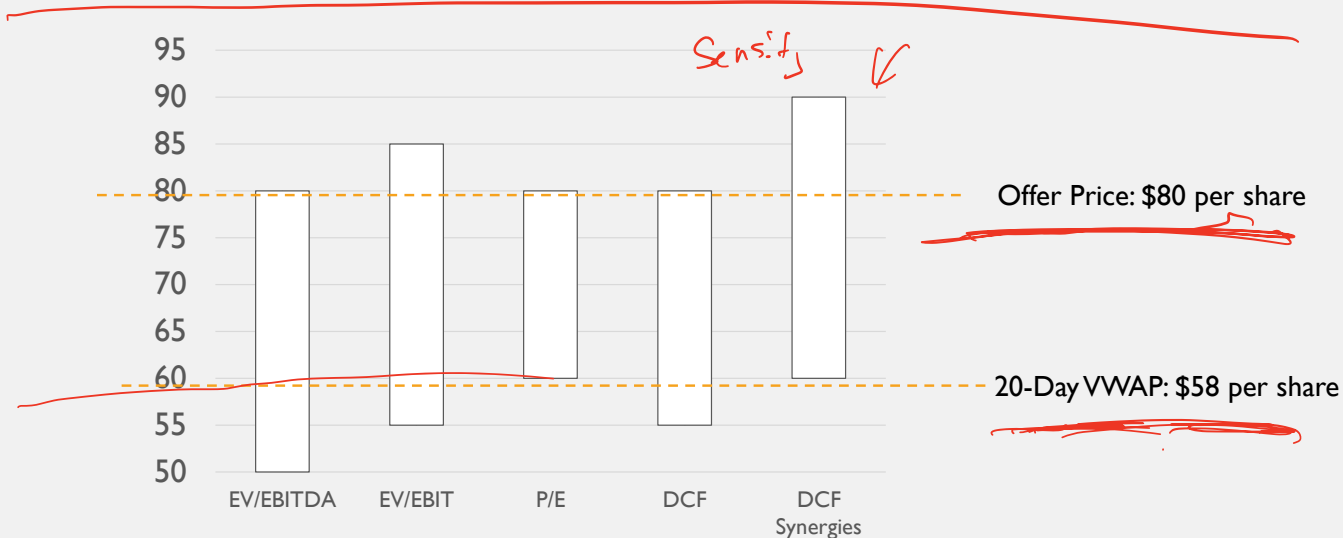
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The "Football Field" (based on arbitrage numbers)

FA Again

The "Football Field" is a graphical summary of valuation results (share value or equity value) based on different methods.



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comparable

high

low

\$80 Not-binding
→ DD

Due Diligence

(Is the target company as good as it appears?)

- Due diligence is the verification and investigation of a potential corporate transaction or investment opportunity including
 - M&As
 - Private equity investment
 - VC investment
- Most M&A failures could be attributed to, at least partially, due diligence failures.



Things to do before due diligence

- What are the acquirer's and the target's major concerns in the due diligence process?
 - Acquirer:
 - Not able to obtain a complete and accurate picture of the target
 - Information leakage might push up the target's share price or attract competing bidders.
 - Target:
 - The acquirer might exploit the confidential information especially when the deal is withdrawn.

customer list

Things to do before due diligence, cont'd

- Prepare a “Non-Disclosure Agreement” (NDA) of due diligence, which typically contains the following items:
 - Whether it is a one-way or two-way sharing of confidential information
 - Definition of “confidential information”
 - Protocol of destroying/returning confidential information if the deal breaks
 - No obligation clause: the acquirer has no obligation to make a binding offer
 - Disclaimer: no guarantee of the absolute accuracy in the presentation of the target’s information

“Data Room”



- All material required for due diligence should be prepared by the target and stored in “Data Room”.
- Most data rooms today are virtual.
 - <https://www3.intralinks.com/>
 - <https://offers.idealsvdr.com/>
 - <https://dealroom.net>

Index	Title	View	Download	Share
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3	Other Contracts and Agreement	View	Download	Share
4	Products and Solutions	View	Download	Share
4.1	Corporate Records	View	Download	Share
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4.3	Labor & Employment	View	Download	Share
4.4	Legal & Other	View	Download	Share
4.5	Market Data	View	Download	Share
4.5.1	Competitive Supply Information.pdf	View	Download	Share
4.5.2	Market Assessment .pdf	View	Download	Share

The sinking of the superyacht “Bayesian” off the coast of Sicily made headlines in many news media websites in 2024

- Source:

- <https://www.theguardian.com/business/article/2024/aug/19/who-is-mike-lynch-tech-boss-missing-yacht-sinking-autonomy>
- <https://www.abc.net.au/news/2024-08-20/what-we-know-about-mike-lynch-bayesian-sinking/104245450>
- <https://www.abc.net.au/news/2025-05-10/diver-dies-in-preliminary-operations-to-recover-lynch-yacht/105277332> ←

- Six people, including Mike Lynch, were killed in this incident. But who is Mike Lynch?

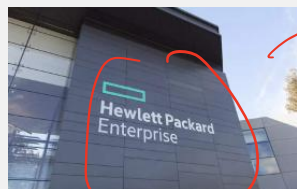


Legal case

I B-L
AUD

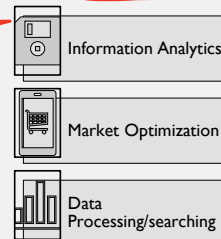
Due diligence example 1:

Hewlett-Packard claimed that it lost more than \$5 bn over the acquisition of Autonomy Corp.



bought Autonomy for \$11.1bn in 2011

then wrote down Autonomy's value by \$8.8bn within a year



Mike
Lynch

Deloitte KPMG

- Source: <https://www.theguardian.com/business/2024/feb/12/hewlett-packard-lost-4bn-over-autonomy-acquisition-london-court-told>

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DD

15

1965
PhD

Discussion

Read the articles on the right and search for additional information if needed.

- <https://www.theguardian.com/business/2024/mar/17/mike-lynch-criminal-trial-british-tech-tycoon>
- <https://www.herbertsmithfreehills.com/notes/corporate/2022-05/directors-found-liable-for-misleading-statements-and-misrepresentations-in-annual-and-quarterly-reports>

Questions:

- What went wrong with HP's acquisition of Autonomy?
- How could the due diligence of the Autonomy acquisition have been improved?

What do the three valuation methods have in common?

1. $FCFF = \text{EBIT} \times (1 - T_c) + D\&A - \Delta NWC - CapEx$

2. $\frac{\text{Enterprise Value}}{\text{EBITDA}}$

$$\frac{EV}{EBIT}$$

3. $\frac{\text{Price}}{EPS}$

What do the three valuation methods have in common?

1. $FCFF = EBIT \times (1 - T_c) + D\&A - \Delta NWC - CapEx$

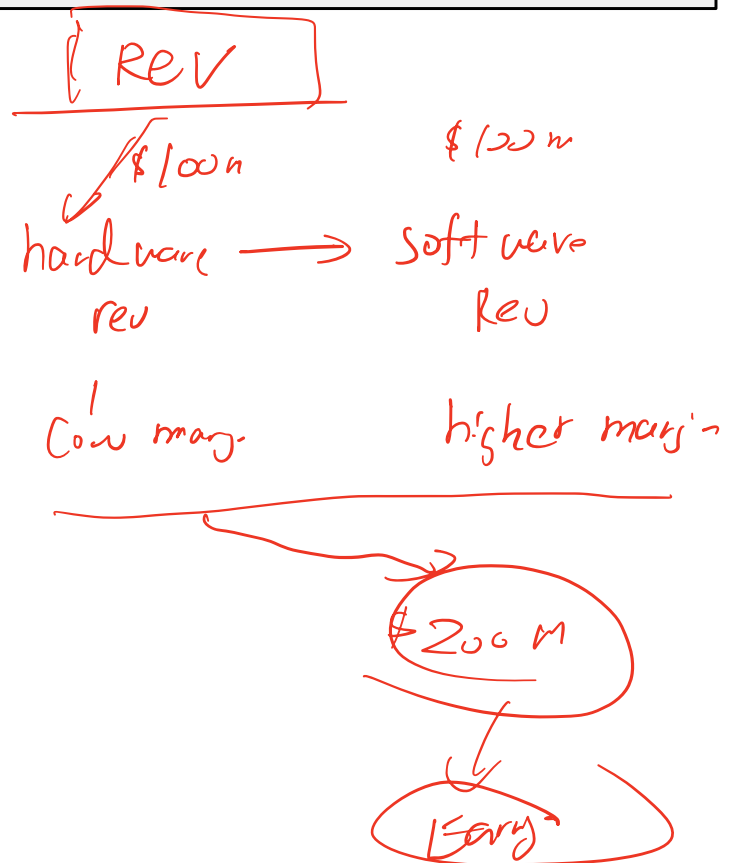
"*earnings* before interest and taxes"

2. $\frac{\text{Enterprise Value}}{EBITDA}$

"*earnings* before interest, taxes, depreciation & amortization"

3. $\frac{\text{Price}}{EPS}$

"*earnings* per share"



Financial due diligence

- A typical starting point of due diligence:
 - Investigate the target's financial statements
- Are there signs of manipulations in the target's financial statements?
 - Compare the target's most recent financial statements with the past financial statements (**trend** analysis)
 - Compare the target's financial statements (most recent and trend) with those of its peers' (**peer** analysis).
- Discussion:
 - What financial figures to check?



In-class due diligence mini case #1

earnings quality

- Poseidon Concepts Corporation (PCC)
 - Your company plans to acquire PCC.
 - Assume the date is September 30, 2012.
 - Review the financial statements of the first three quarters of 2012.
 - Q1: Are there any unusual patterns in the **trend**?
 - Q2: If yes, what problems do they imply?
 - Q3: Any other signals of hidden problems of the target?

① NI doubled ??

② mgt offload

③ AIR ↑

Summary of due diligence on the quality of earnings

- The typical starting point of due diligence: the **quality of earnings**:

positives
earnings

- Investigate accounting **accruals** (net working capital items such as accounts receivables)
- trend** analysis and **peer** comparison
 - An unjustified increase in accounts receivables could mean manipulated revenues.
 - An unjustified increase in accounts payables could mean manipulated expenses.

Automatic
↓

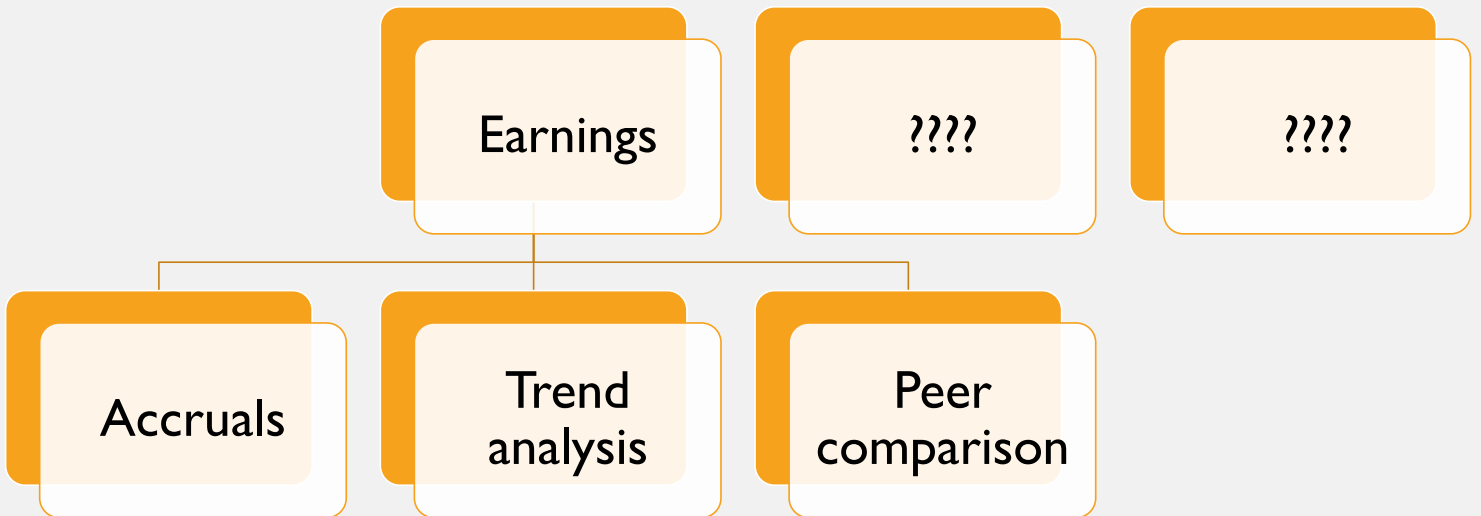
- Other financial statement items to check:

- Unusual change in debt and unusual dividend payments.

Hard → Software
Rev Rev
└──────────┘

- Question: what if the target firm is in a very early stage and does not have positive earnings?

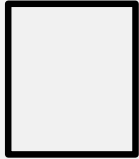
What else to check in due diligence?



Have a guess which company has this board?



1) CEO



2) President



3) **Sam Nunn**,
former Senator



6) **Gary Roughhead**,
former Admiral, US Navy



9) **Bill Foege**,
Former Director, CDC



4) **George Schultz**,
former Secretary of State



7) **Henry Kissinger**,
former Secretary of State



10) **Riley Bechtel**,
Bechtel Inc.



5) **Bill Perry**,
former Secretary of Defense



8) **Dick Kovacevich**, former
CEO, Wells Fargo



11) **James Mattis**,
former Marine Corps General

Therona

Due diligence example 2: Theranos

- 2003: Theranos, a private biotech company was founded by 19-year-old Elizabeth Holmes.
 - The company claimed that it had invented a breakthrough blood-testing method.
- 2014: the company was valued at \$9 bln after several rounds of funding.

- Major investors:



- Sources: <https://www.dailymail.co.uk/news/article-12092549/Theranos-fraudster-Elizabeth-Holmes-ordered-repay-Rupert-Murdoch-entire-125-MILLION-invested.html>

Example 2: Theranos, cont'd

- 2015: A WSJ article questioned the biotech company's technology.
- 2018: SEC charged the company's founder (Elizabeth Holmes) with massive fraud.
- 2022: Holmes was sentenced to 11 years.
- 2023: Holmes reported to federal prison camp Bryan in southern Texas.

Hot Startup Theranos Has Struggled With Its Blood-Test Technology

Silicon Valley lab, led by Elizabeth Holmes, is valued at \$9 billion but isn't using its technology for all the tests it offers

By John Carreyrou [Follow](#)

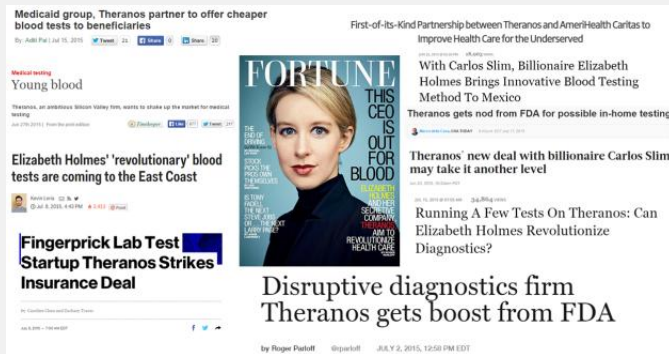
Updated October 16, 2015

- Sources:
 - <https://www.wsj.com/articles/theranos-has-struggled-with-blood-tests-1444881901>
 - <https://www.investopedia.com/articles/investing/020116/theranos-fallen-unicorn.asp>

In-class Discussion

due diligence of assets

- Assume we are in early 2015.
- Take the role of a major biotech company's CEO.
- You are interested in acquiring Theranos in early 2015.



Questions:

1. What types of due diligence activities need to be undertaken before the acquisition/investment?
2. What challenges might arise during the due diligence process for Theranos?

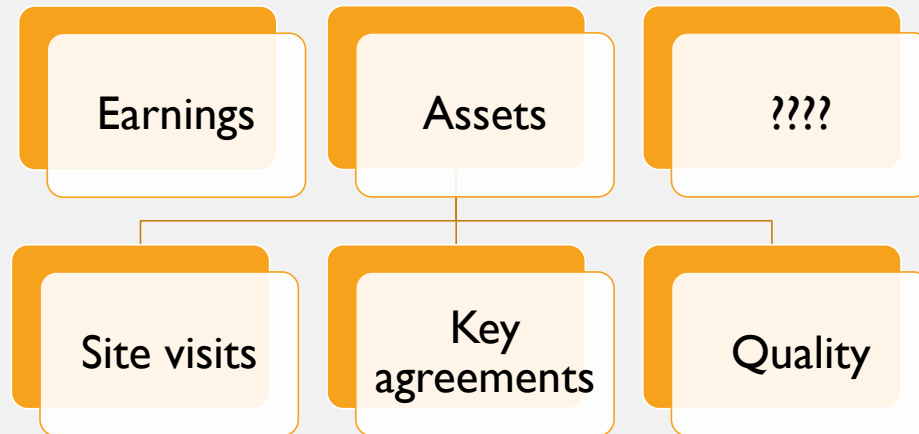
For more information:

- <https://www.cbinsights.com/research/theranos-news-sentiment/>
- <https://www.wired.com/story/what-biotech-ceos-and-their-funders-learned-from-the-theranos-debacle/>

Due diligence: Assets

- | | |
|---|---|
| <ul style="list-style-type: none">• Physical assets:<ul style="list-style-type: none">• Property, Plant, and Equipment (PP&E)• Inventories• What DD activities/concerns are most relevant for physical assets?<ul style="list-style-type: none">• Site visits• Borrowing agreements: whether these assets were used as collateral.• | <ul style="list-style-type: none">• Intellectual properties:<ul style="list-style-type: none">• Copyrights• Patents• Trademarks• Digital assets• What DD activities/concerns are most relevant for IPs?<ul style="list-style-type: none">• Ownership of the IP• Key agreements involving the IP (e.g., whether it has been used in any joint ventures with other companies)• The ability of the IP to generate cash flows• |
|---|---|

Due diligence: Assets



Due diligence: Hidden Liabilities/Issues

E.g. of other items on the DD check list	Reasons
Employment contracts (including those of executives)	Employment contracts include redundancy payout and benefits (e.g., a “golden parachute”)
Pending lawsuits	The acquirer may inherit significant liabilities/fees/penalties if it loses the lawsuits in the future.
Product warranties	Future customer claims and costs.
Unusually high professional fees	It may suggest hidden issues.
Unusual stock transactions of the top management	It may suggest hidden issues.
Unusual high turnovers of the top management	It may suggest hidden issues.